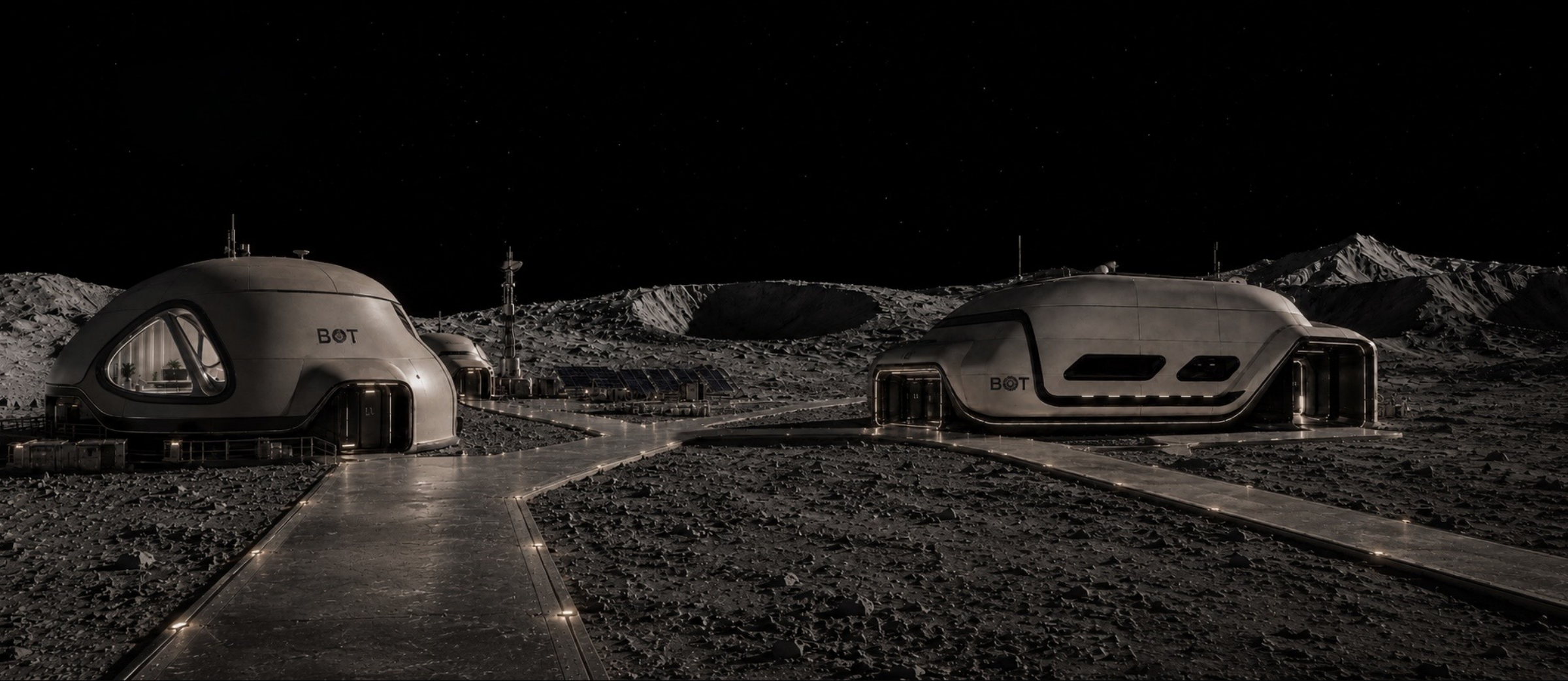


Moontology

A world built to be understood.



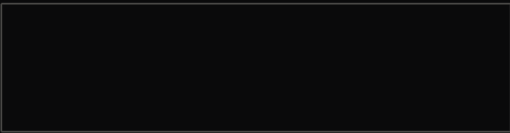
ARCHIVED GENERATION / CONNECTED HABITATS & WALKWAY

Mission Success

A near-future lunar worksite where a robot can inspect what other machines claim to be doing. Moontology joins a World Labs environment, articulated agents and an ontology of assets, tasks and resources so an operator can see a discrepancy, understand its consequences and choose the next action.

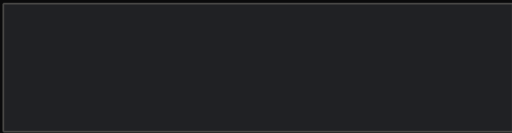


COLOR ANCHORS
Optimus / Cybercab



SENSOR BLACK
#0A0A0B

Tight specular highlights.
Visors, shells, sensing apertures.



GRAPHITE
#202124

Broad, subdued reflection.
Joints and secondary structures.



OPTIMUS CHAMPAGNE
#BFB1A0

Soft satin metal.
Selective panels and hardware.



CYBERCAB CHAMPAGNE
#AB8F74

Deeper satin warmth.
Junctions and protective trim.

OPERATING PREMISE

2030 is the design horizon: a base still being built. Excavation, cargo, power distribution and a dust-managed threshold give the scene its jobs, routes and reasons to exist. NASA’s phased infrastructure roadmap supplies the context. [1]

DESIGN RULE

Gloss reveals sensing surfaces. Champagne marks selected hardware. Graphite quiets the mechanisms.

[1] NASA Moon Base phases

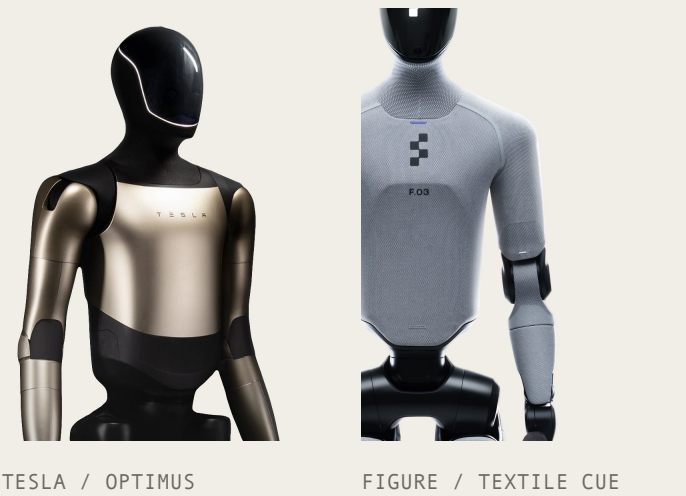
Palette: original design record

Project color standards; image-derived references, not Tesla paint specifications.

A considered anatomy.

Technical footwear, a continuous sensor face and selective metal accents were resolved through a sequence of explicit revisions.

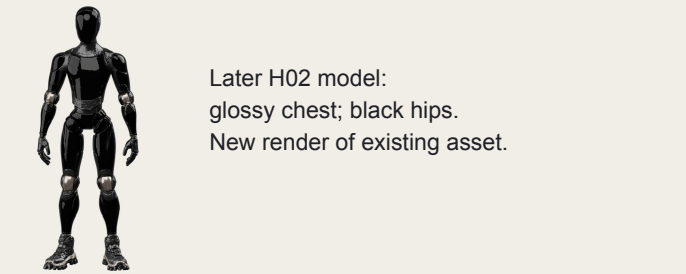
SOURCE REFERENCES



Helmet contour and perimeter light meet a softer textile vocabulary at articulation zones. The brief explicitly removed branding.



A contained upper, technical closure and pronounced outsole establish a terrain-oriented foot. The zipper was then specified diagonally across the front, with champagne confined to its line.



AN ARCHIVE OF DECISIONS



- 1 SENSOR FACE
A single dark face with a fine contour light: orientation without an expressive face.
- 2 FINISH HIERARCHY
Mesh was tested on chest and inner thighs. Later edits returned chest and neck to shiny black.
- 3 WRIST-LEVEL EVIDENCE
Small wrist cameras and matte hand accents place sensing close to the work.
- 4 A CLEANER SILHOUETTE
The champagne groin panel was explicitly removed; black reconnects the torso and legs.

ARCHIVED GENERATION / DESIGN SHEET

Original image used for 3D development.
Earlier finish study; later revisions at left.

01 / TEXTILE STUDY

“Make the chest plate fabric mesh”

02 / FINISH REVISION

“...neck shiny black”

03 / SILHOUETTE EDIT

“remove the gold on the groin hip area and make it black”

Short excerpts from the original revision prompts. Footwear is a visual translation; lunar traction, sealing and thermal performance were not validated.

Source: Generate Humanoid Asset · uploaded references and revision sequence

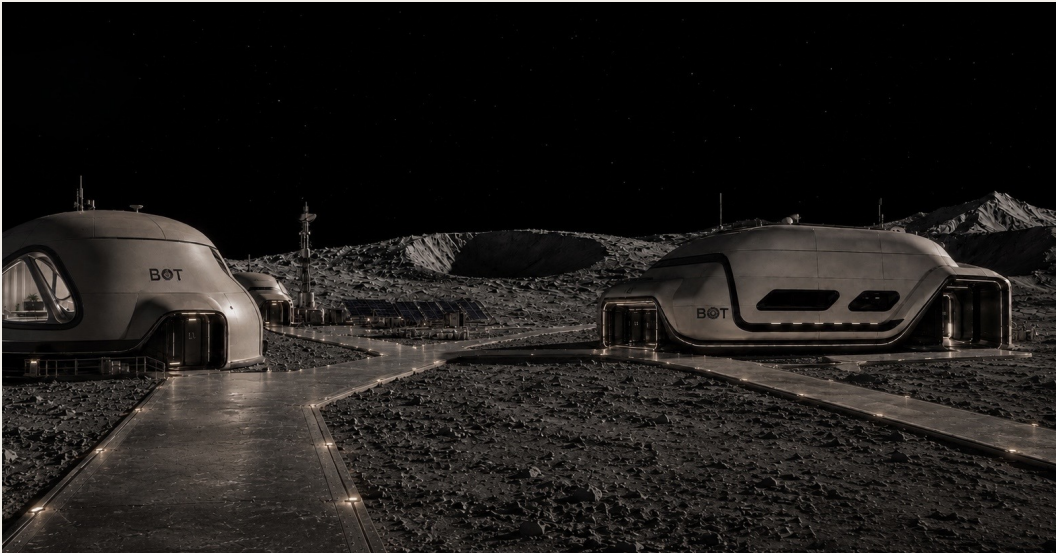
From precedent to place.

Distributed infrastructure becomes connected habitat massing; architectural spectacle is edited into an operational interior.



NASA / PHASE TWO CONCEPT

Distributed mobility, power and logistics provide the program. GRU’s lunar-building research adds a construction reference, rather than a literal hotel replica. [1, 2]



ARCHIVED GENERATION / CONNECTED HABITATS

The user selected the dome, extended the hatch and connected the walkway. Low massing, tinted apertures and open terrain reserve room for independent actors.



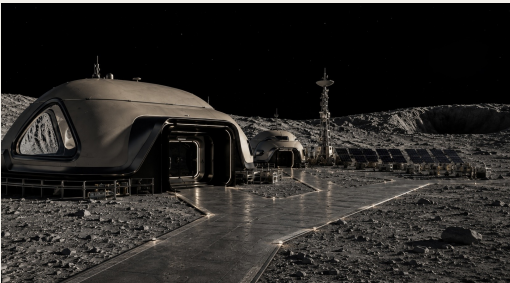
LUCAS MUSEUM / USER-SUPPLIED ARCHITECTURE

Ma Yansong / MAD’s flowing structure and tubular lift core provide the visual reference. The comparison makes the borrowed spatial cues visible. [3]



ARCHIVED GENERATION / INTERIOR

White curves become warm metal; the central core and arch rhythm remain. Subsequent prompts add service utility and tune lighting, flooring and scale to the exterior.



A / FACADE



B / EXTENSION



C / PANORAMA



D / THRESHOLD

Generation rounds: selected facade + walkway → left/right expansion → four connected view requests → SPZ + collider exports. Stitching errors prompted correction; Earth was separated conceptually to avoid duplicate planets.

[1] NASA [2] GRU [3] Lucas Museum Original architecture & map revisions

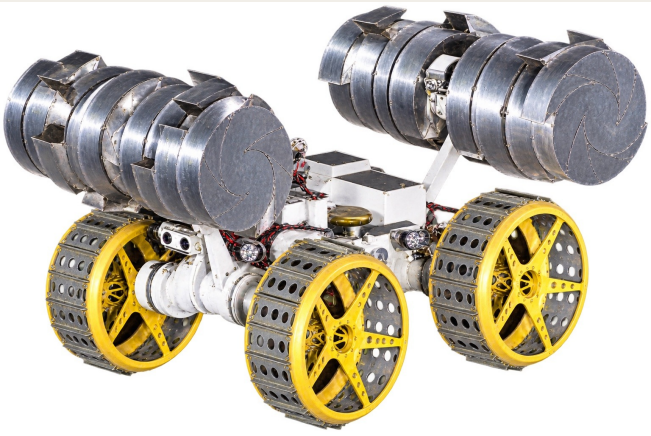
Keep the purpose. Edit the language.

Some forms are preserved almost literally. Others are recast as members of a common black-and-champagne product family.



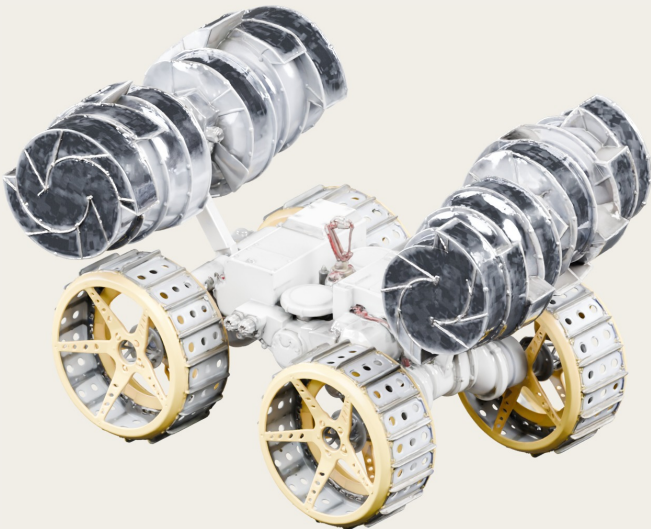
NASA IPEX / UPLOADED REFERENCE

The exact project link is NASA's Infrastructure Pilot Excavator. Opposed bucket drums counter excavation reaction forces in reduced gravity. [1]



ARCHIVED GENERATION

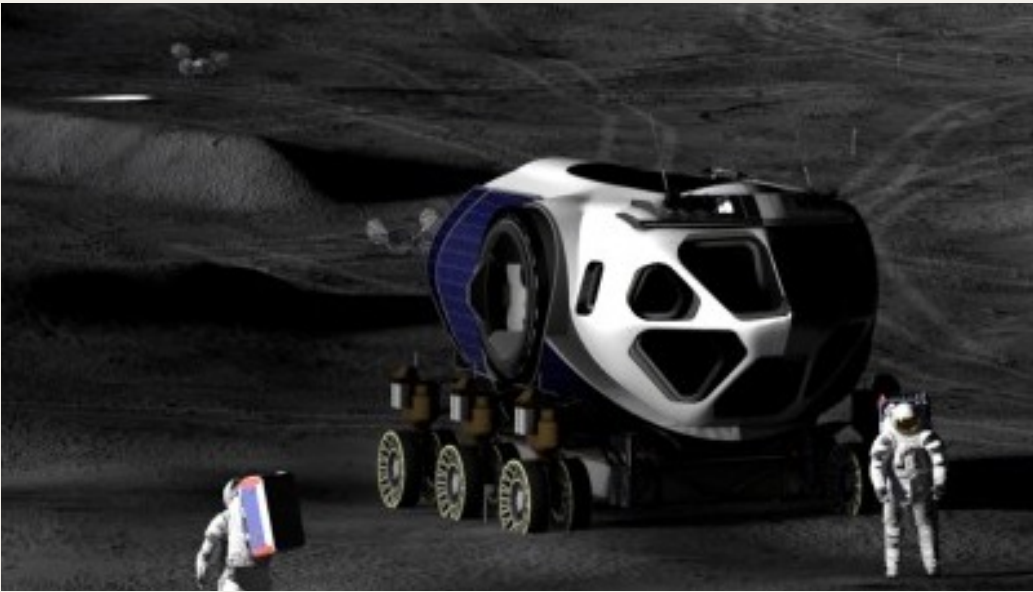
The lunar reference becomes an isolated asset. Opposite-side and top-down rounds correct the relation between the drums and wheels.



EXISTING 3D ASSET / NEW RENDER

The retained mesh still carries silver/yellow source finishes. Its hardware lineage is direct; palette unification is incomplete.

Revision excerpt: “Align them parallel with the positioning of the tires”

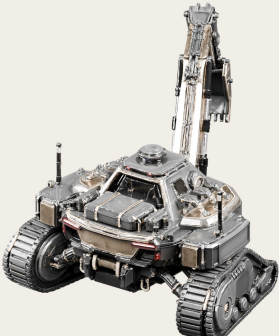


NASA / PRESSURIZED MOBILITY PRECEDENT

The faceted window pattern, side hatch and multi-wheel stance survive. White/blue surfaces become champagne with dark glazing and restrained highlights; the recognizable vehicle remains the anchor. [2]



ARCHIVED GENERATION / MATERIAL TRANSLATION



ARTICULATED EXCAVATOR / 3D



CABLE DEPLOYMENT / CONCEPT



CARGO MOBILITY / CONCEPT

[1] NASA IPEX + original asset brief

Excavator conversation

[2] NASA Phase Two

Vehicle palette & cable prompt

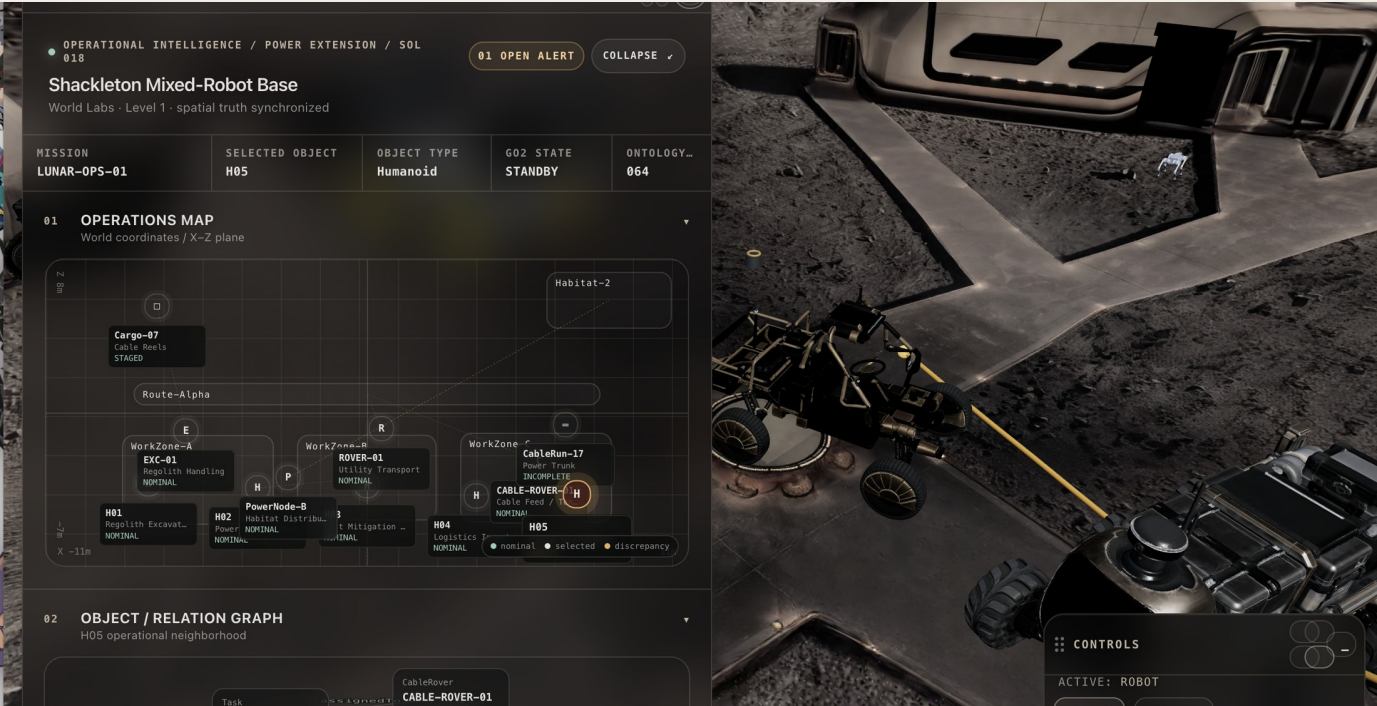
Appearance becomes evidence.

The design ends with a useful distinction: what should happen, what an asset reports, and what an independent observer can verify.



UNITREE / OFFICIAL URDF + DAE MESHES

Imported from unitree_ros, revision 7d6075f. Geometry is inherited, not generated. This new posed render exposes the official articulated chain. [1]



ARCHIVED THREE.JS RUNTIME / 04 SEPTEMBER

INTEGRATION CONTRACT

WORLD LABS

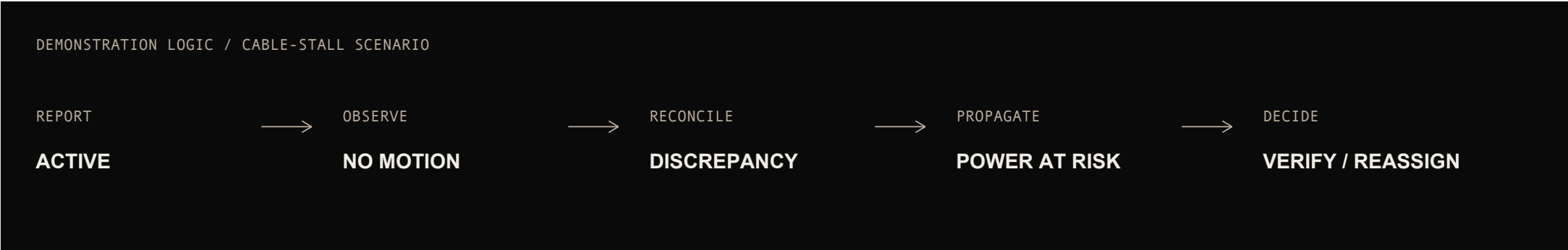
SPZ appearance + collider mesh. Exterior and interior are separate worlds.

THREE.JS

Independent assets; AgentRoot owns pose. URDF/VisualRig provides articulation.

ONTOLOGY

Entities, task dependencies, observations and actions share one typed state model.



The cable-stall chain is the documented scenario. Current source retains expected/reported/observed records and verification actions; the screenshot is a development capture, not proof of a completed physical mission.

NASA / MOON BASE

Power distribution, cargo, regolith and dust-tolerant connections are explicit needs. The ontology gives each a state and dependency. Guide p.13.

NASA JPL / CADRE

Cooperating robots share measurements and act with reduced ground intervention. Moontology explores independent inspection across a fleet.

NIST / DIGITAL TWIN CORE

Common information models and interoperable services motivate stable asset identity, evidence provenance and one canonical state.

SPACE FORCE / DIGITAL SERVICE

Interconnected models and data-informed operations support the case for traceable state and operator decisions. A cross-domain analogy.

DIU / AISUM

Multi-vendor autonomous teams and common communications support modular adapters and single-operator fleet supervision. A terrestrial precedent.

Program relevance is an argued fit, not certification, procurement status or agency endorsement. Lunar environmental qualification and end-to-end physical autonomy remain separate validation work.

[1] Official Unitree source

Ontology design record

NASA architecture guide